

Traffic Signaling

Sub-techniques (2)

Adversaries may use traffic signaling to hide open ports or other malicious functionality used for persistence or command and control. Traffic signaling involves the use of a magic value or sequence that must be sent to a system to trigger a special response, such as opening a closed port or executing a malicious task. This may take the form of sending a series of packets with certain characteristics before a port will be opened that the adversary can use for command and control. Usually this series of packets consists of attempted connections to a predefined sequence of closed ports (i.e. [Port Knocking](#)), but can involve unusual flags, specific strings, or other unique characteristics. After the sequence is completed, opening a port may be accomplished by the host-based firewall, but could also be implemented by custom software.

Adversaries may also communicate with an already open port, but the service listening on that port will only respond to commands or trigger other malicious functionality if passed the appropriate magic value(s).

The observation of the signal packets to trigger the communication can be conducted through different methods. One means, originally implemented by Cd00r [\[1\]](#), is to use the libpcap libraries to sniff for the packets in question. Another method leverages raw sockets, which enables the malware to use ports that are already open for use by other programs.

On network devices, adversaries may use crafted packets to enable [Network Device Authentication](#) for standard services offered by the device such as telnet. Such signaling may also be used to open a closed service port such as telnet, or to trigger module modification of malware implants on the device, adding, removing, or changing malicious capabilities. Adversaries may use crafted packets to attempt to connect to one or more (open or closed) ports, but may also attempt to connect to a router interface, broadcast, and network address IP on the same port in order to achieve their goals and objectives.[\[2\]\[3\]\[4\]](#) To enable this traffic signaling on embedded devices, adversaries must first achieve and leverage [Patch System Image](#) due to the monolithic nature of the architecture.

Adversaries may also use the Wake-on-LAN feature to turn on powered off systems. Wake-on-LAN is a hardware feature that allows a powered down system to be powered on, or woken up, by sending a magic packet to it. Once the system is powered on, it may become a target for lateral movement.[\[5\]\[6\]](#)

ID: T1205

Sub-techniques: [T1205.001](#), [T1205.002](#)

 Tactics: [Stealth](#), [Persistence](#), [Command and Control](#)

 Platforms: Linux, Network Devices, Windows, macOS

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[Version Permalink](#)

Procedure Examples

ID	Name	Description
S9011	BRUSHFIRE	BRUSHFIRE has monitored inbound VPN traffic to compromised appliances until specific inbound packets contain a specific magic string/pattern instead of external beaconing. ^[7]
S1118	BUSHWALK	BUSHWALK can modify the <code>DSUserAgentCap.pm</code> Perl module on Ivanti Connect Secure VPNs and either activate or deactivate depending on the value of the user agent in incoming HTTP requests. ^[8]
S0220	Chaos	Chaos provides a reverse shell is triggered upon receipt of a packet with a special string, sent to any port. ^[9]
C0029	Cutting Edge	During Cutting Edge , threat actors sent a magic 48-byte sequence to enable the PITSOCK backdoor to communicate via the <code>/tmp/clientsDownload.sock</code> socket. ^[8]
S1203	J-magic	J-magic can monitor TCP traffic for packets containing one of five different predefined parameters and will spawn a reverse shell if one of the parameters and the proper response string to a subsequent challenge is received. ^[10]
G0094	Kimsuky	Kimsuky has used TRANSLATEX to redirect clients to legitimate Gmail, Naver or Kakao pages if the clients connect with no parameters. ^[11]
S0641	Kobalos	Kobalos is triggered by an incoming TCP connection to a legitimate service from a specific source port. ^{[12][13]}
S9043	Mini Shai-Hulud	Mini Shai-Hulud has examined commit messages for a keyword followed by base64 encoded segments to validate communications and to execute subsequent actions to include exfiltration. ^[14]
G0129	Mustang Panda	Mustang Panda has utilized a magic value in C2 communications and only executes in memory when response packets match specific values of "17 03 03" or "46 77 4d". ^[15]
S0664	Pandora	Pandora can identify if incoming HTTP traffic contains a token and if so it will intercept the traffic and process the received command. ^[16]
S0587	Penguin	Penguin will connect to C2 only after sniffing a "magic packet" value in TCP or UDP packets matching specific conditions. ^{[17][18]}
S1228	PUBLOAD	PUBLOAD has utilized a magic value in C2 communications and only executes in memory when response packets match specific values of 17 03 03. ^{[15][19][20][21][22]} PUBLOAD has also used magic bytes consisting of 46 77 4d. ^[15]
C0056	RedPenguin	During RedPenguin , UNC3886 leveraged malware capable of inspecting packets for a magic-string to activate backdoor functionalities. ^[23]
S1219	REPTILE	The REPTILE reverse shell component can listen for a specialized packet in TCP, UDP, or ICMP for activation. ^{[24][25]}
S0446	Ryuk	Ryuk has used Wake-on-Lan to power on turned off systems for lateral movement. ^[5]
S0519	SYNful Knock	SYNful Knock can be sent instructions via special packets to change its functionality. Code for new functionality can be included in these messages. ^[3]
S1239	TONESHELL	TONESHELL has utilized a magic value in C2 communications and only executes in memory when response packets match specific values. ^{[26][21][27]}
S1201	TRANSLATEX	TRANSLATEX has redirected clients to legitimate Gmail, Naver or Kakao pages if the clients connect with no parameters. ^[11]
S0221	Umbreon	Umbreon provides additional access using its backdoor Espeon, providing a reverse shell upon receipt of a special packet. ^[28]

ID	Name	Description
S0430	Winnti for Linux	Winnti for Linux has used a passive listener, capable of identifying a specific magic value before executing tasking, as a secondary command and control (C2) mechanism. ^[30]
S1114	ZIPLINE	ZIPLINE can identify a specific string in intercepted network traffic, <code>SSH-2.0-OpenSSH_0.3xx.</code> , to trigger its command functionality. ^[31]

Mitigations

ID	Mitigation	Description
M1042	Disable or Remove Feature or Program	Disable Wake-on-LAN if it is not needed within an environment.
M1037	Filter Network Traffic	Mitigation of some variants of this technique could be achieved through the use of stateful firewalls, depending upon how it is implemented.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0524	Traffic Signaling (Port-knock / magic-packet → firewall or service activation) – T1205	AN1448	A remote host sends a short sequence of failed connection attempts (RST/ICMP unreachable) to a set of closed ports. Within a brief window the endpoint (a) adds/enables a firewall rule or (b) a sniffer-backed process begins listening or opens a new socket, after which a successful connection occurs. Also detects Wake-on-LAN magic packets seen on local segment.
		AN1449	Closed-port knock sequence from a remote IP followed by on-host firewall change (iptables/nftables) or daemon starts listening (socket open) and a successful TCP/UDP connect. Optional detection of libpcap/raw-socket sniffers spawning to watch for secret values.
		AN1450	Remote knock sequence followed by PF/socketfilterfw rule update or a background process listening on a new port; then a successful TCP session. Also flags WoL magic packets on local segment.
		AN1451	Crafted ‘synful knock’ patterns toward routers/switches (same src hits interface/broadcast/network address on same port in short order) followed by ACL/telnet/SSH enablement or module change. Detect device image/ACL updates then a new mgmt session.

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